



Cape Peninsula
University of Technology

creating futures

GUIDELINES FOR EMPLOYERS

Diploma in Information and Communication Technology: Applications Development

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1. INTRODUCTION

Applications Development is one of three specialisation areas in the Diploma in Information and Communication Technology offered by the Department of IT at CPUT.

The purpose of the ICT Diploma in Applications Development is to acquire sufficient knowledge to solve complex problems in an ethical and professional manner by obtaining, processing, evaluating, managing and communicating related information using a range of suitable methods within the area of study as indicated by the different specialization purpose statements.

It is CPUT's goal that almost all learners will have had some exposure to the world of work before they graduate. This is done through the Work Integrated Learning (WIL) programmes.

2. WHAT IS WORK INTEGRATED LEARNING (WIL)?

The Work Integrated Learning Research Unit (WILRU), based at the Cape Peninsula University of Technology (CPUT), has preferred to use the term Work Integrated Learning (WIL) to describe an approach to career-focused education. WIL programmes at CPUT are aimed at producing well-grounded and socially responsive graduates in partnership with business, government, community and other higher education institutions.

There are 4 curricular types that are implemented at CPUT which are:

- Work directed theoretical learning
- Problem based learning
- Project based learning
- Workplace learning

The purpose of the curricular types listed above is to support transfer and development of knowledge learnt at university to workplace situations. Through recontextualisation, knowledge learned in one site (the university) is transferred to another (the workplace). This follows a 5 step process:

- Understanding the new context
- Initial extraction of academic elements
- Identifying learnt skill relevant to the work situation
- Transforming knowledge for the new context
- Integration into a new situation.

3. PROFILE OF AN APPLICATIONS DEVELOPMENT INTERN

The incumbent is a technical, creative and responsible person capable of functioning alone or as part of a team. The Intern has a broad and solid foundation in applications development with an aptitude to easily adapt to a new ICT environment if so required.

Interns are not limited to industry exposure but may also be exposed to community engagement for social upliftment. The Intern is expected to be an active participant in the industry project(s) and to use their skills (and develop further skills) to assist the organisation reach its goals.

4. CURRICULUM AND INDUSTRY SKILLS ASSESSMENT CRITERIA

The purpose of the ICT Diploma in Applications Development is to acquire sufficient knowledge to solve complex problems in an ethical and professional manner by obtaining, processing, evaluating, managing and communicating related information using a range of suitable methods within the area of study as indicated by the different specialization purpose statements.

In the First year of study, all students enrol for the same subjects. Thereafter, in the second year of study a student is required to choose their preferred specialization area. The Diploma in ICT qualification has three specialisation areas, each with its own purpose statement, compulsory and elective subjects. The three specialisation areas are Applications Development, Communication Networks and Multimedia Technology.

The subjects are as follows:

Level 1	Level 2	Level 3
Applications Development Foundations 1 Business Practice 1 Communications Networks Foundations 1 ICT Fundamentals 1 Multimedia Foundations 1 Professional Communications 1 Programming 1 Project 1	Applications Development Fundamentals 2 Applications Development Practice 2 Applications Development Theory 2 Communications Networks Fundamentals 2 ICT Electives 2 Information Management 2 Information Systems 2 Multimedia Applications Fundamentals 2 Professional Communications 2 Project 2	Applications Development Practice 3 Applications Development Theory 3 ICT Electives 3 Information Systems 3 Professional Practice 3 Project 3 Project Management 3 Project Presentation 3

Interns would benefit from exposure to any or a combination of the 'basket of competencies' listed below:

- Database programming (SQL scripting, stored procedures, etc)
- Web design and implementation
- Web programming
- Client-side scripting
- Server-side programming
- Development using a CMS like Wordpress or Joomla
- Code management using GIT or similar
- Search Engine Optimisation
- Domain-Driven design
- Implementation of (GOF) patterns
- Using frameworks like Spring
- Exposure to different programming languages not done as part of the current curriculum
- Mobile programming using Android/iOS
- Game design and development
- TDD (Test Driven Development)

5. RESPONSIBILITIES OF THE EMPLOYER

Our aim is always to develop healthy working relationships between industry, the learner and the institution.

It is the employer's responsibility:

- To assist in providing the intern with meaningful employment related to their field of study
- To familiarise the incoming intern with the employer's expectations
- To discuss the intern's training plan at the beginning of the training period
- To provide supervision and training to the intern by individuals who understand and are interested in co-operative education
- To evaluate the intern's work using the logbook on a weekly basis
- To provide an overall evaluation of the intern's competency at the end of the experiential training period.

6. RESPONSIBILITIES OF THE INTERN

The intern is required to always behave in a professional manner and to avoid any action that may bring CPUT or the Employer in any disrepute whatsoever. Interns should not deviate from the listed duties without consent from BOTH the Employer and CPUT.

Interns are expected to:

- Register for WIL (it is compulsory)
- Ensure that employer details are forwarded to the WIL coordinator
- Conform to company policies and procedures
- Follow safety rules explicitly
- Conduct themselves professionally and ethically throughout the application and placement process as well as on the job
- Contact the WIL coordinator immediately in the event of queries or dissatisfaction concerning the workplace learning programme
- Ensure that the employer/supervisor evaluate the logbooks on a weekly basis
- Submission of signed copies of the evaluated log sheet to the respective WIL coordinator on a monthly basis via Blackboard
- To prepare and collect the necessary documentation, reports required for workplace learning evaluation
- Identify and document learning experiences and skills obtained
- To gather and create evidence (appropriate items of proof) of learning experiences, proving competence and skills
- To complete a portfolio that demonstrates proficiency in several criteria to be presented on returning to CPUT

7. RESPONSIBILITIES OF CPUT

It is the responsibility of the university to place students, through the faculty/department designated WIL Coordinator, within suitable work integrated learning programmes. Staff within the faculty, through the WIL

coordinator, are responsible for establishing guidelines for the workplace experience and the monitoring and evaluation thereof.

8. DURATION

The work integrated learning module forms part of the Project Presentation subject and is compulsory. The duration of this module is six months. Interns will attend lectures on campus every Monday and work full time from Tuesday to Friday. They will not qualify for the credits for this module if they fail to complete the entire period.

9. CONTACT DETAILS OF RESPONSIBLE PERSONS

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